

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
```

```
df = pd.read_csv("medical_dataset.csv")
```

```
df
```

	Age	BMI	BloodPressure	Disease
0	58	39.3	104	1
1	36	22.7	169	0
2	69	39.6	159	1
3	75	33.1	174	1
4	42	23.7	159	0
..	...	...	...	...
145	57	38.1	108	1
146	58	38.8	148	1
147	53	21.6	107	0
148	77	33.7	136	1
149	39	34.5	120	0

```
[150 rows x 4 columns]
```

```
df.head()
```

	Age	BMI	BloodPressure	Disease
0	58	39.3	104	1
1	36	22.7	169	0
2	69	39.6	159	1
3	75	33.1	174	1
4	42	23.7	159	0

```
df.tail()
```

	Age	BMI	BloodPressure	Disease
145	57	38.1	108	1
146	58	38.8	148	1
147	53	21.6	107	0
148	77	33.7	136	1
149	39	34.5	120	0

```
df.shape
```

```
(150, 4)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Age             150 non-null   int64
1   BMI             150 non-null   float64
2   BloodPressure   150 non-null   int64
3   Disease         150 non-null   int64
dtypes: float64(1), int64(3)
memory usage: 4.8 KB
```

```
df.isnull().sum()
```

```
Age             0
BMI             0
BloodPressure    0
Disease         0
dtype: int64
```

```
df.describe()
```

	Age	BMI	BloodPressure	Disease
count	150.000000	150.000000	150.000000	150.000000
mean	50.660000	29.294667	136.653333	0.493333
std	16.327944	6.816106	25.705087	0.501630
min	20.000000	18.000000	90.000000	0.000000
25%	37.250000	23.625000	113.750000	0.000000
50%	52.500000	29.600000	136.000000	0.000000
75%	62.750000	35.400000	160.000000	1.000000
max	79.000000	39.700000	178.000000	1.000000

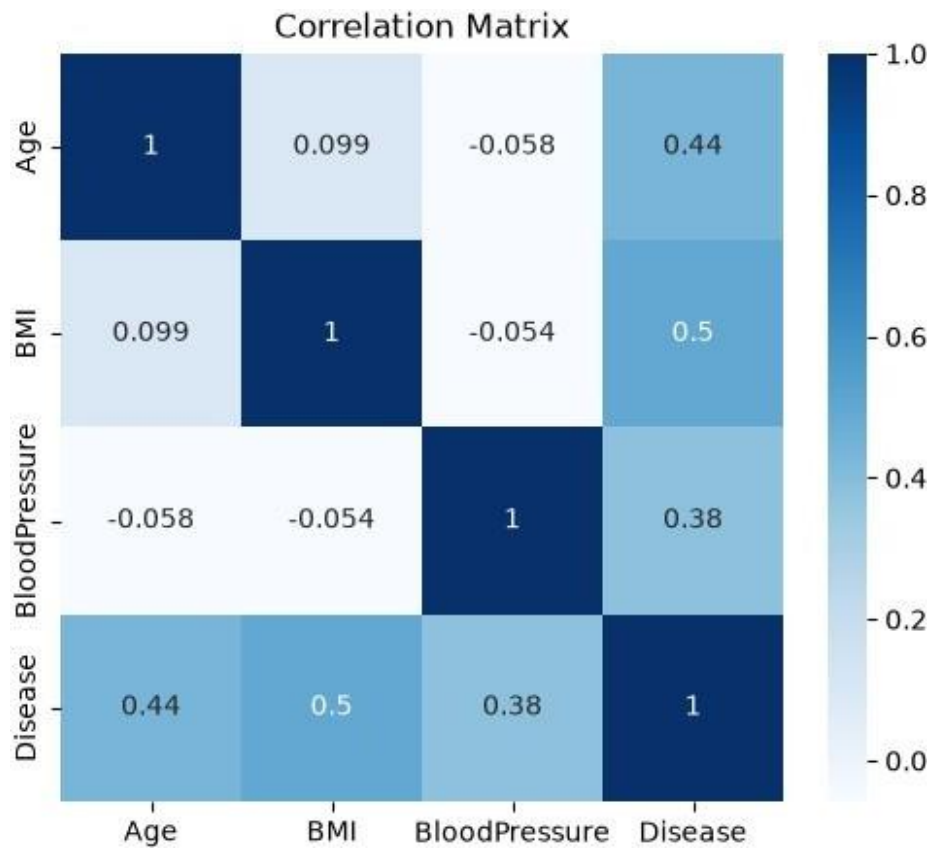
```
df.duplicated().sum()
```

```
np.int64(0)
```

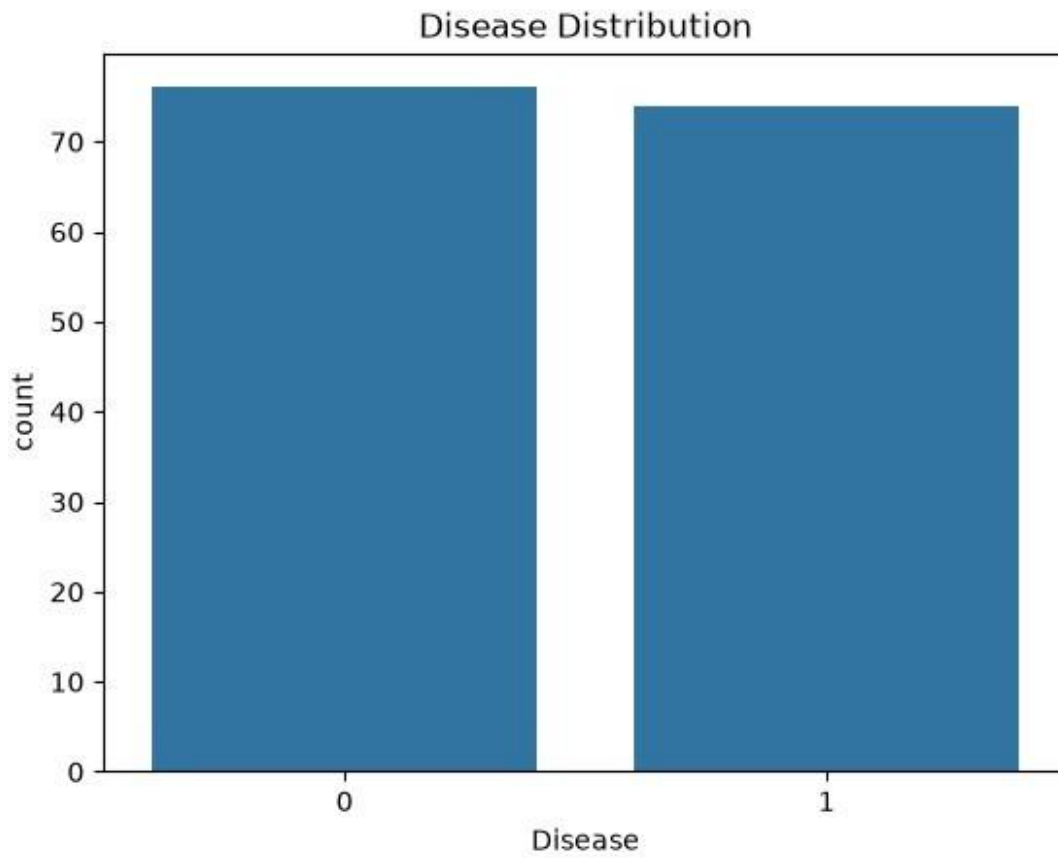
```
df.corr()
```

	Age	BMI	BloodPressure	Disease
Age	1.000000	0.099310	-0.057513	0.435235
BMI	0.099310	1.000000	-0.053841	0.496794
BloodPressure	-0.057513	-0.053841	1.000000	0.383419
Disease	0.435235	0.496794	0.383419	1.000000

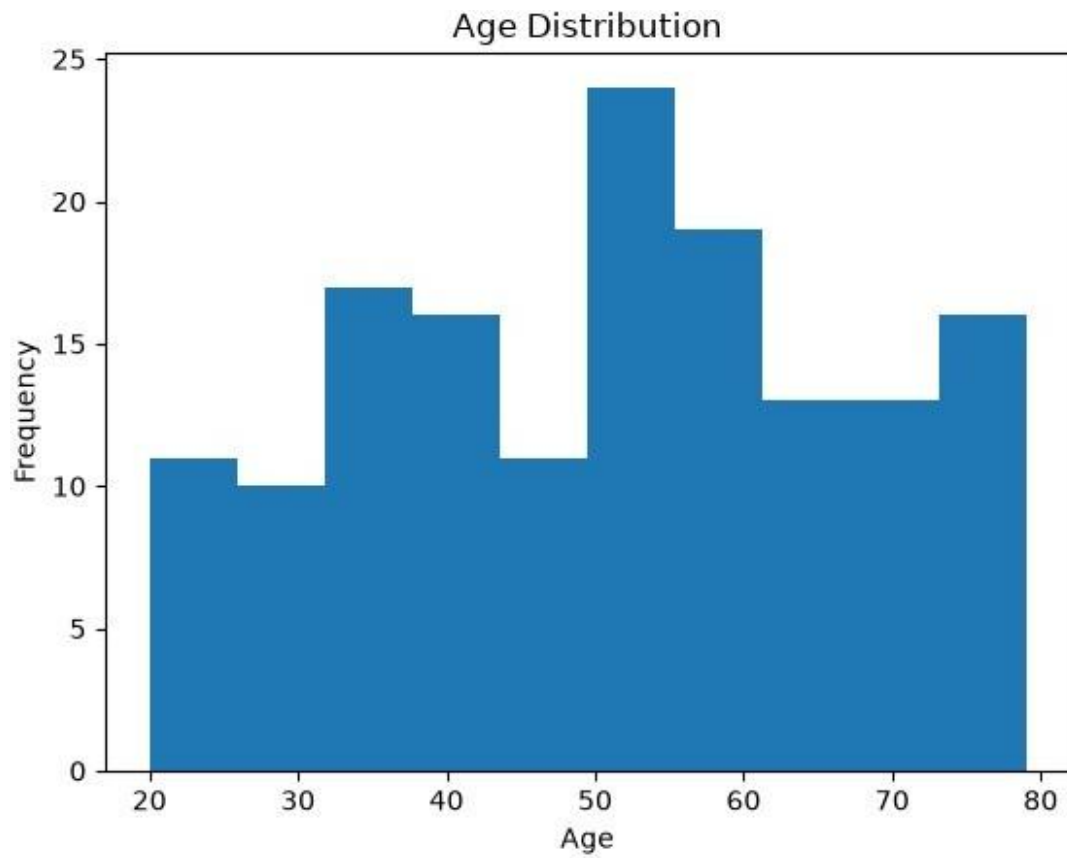
```
plt.figure(figsize=(6,5))
sns.heatmap(df.corr(), annot=True, cmap="Blues")
plt.title("Correlation Matrix")
plt.show()
```



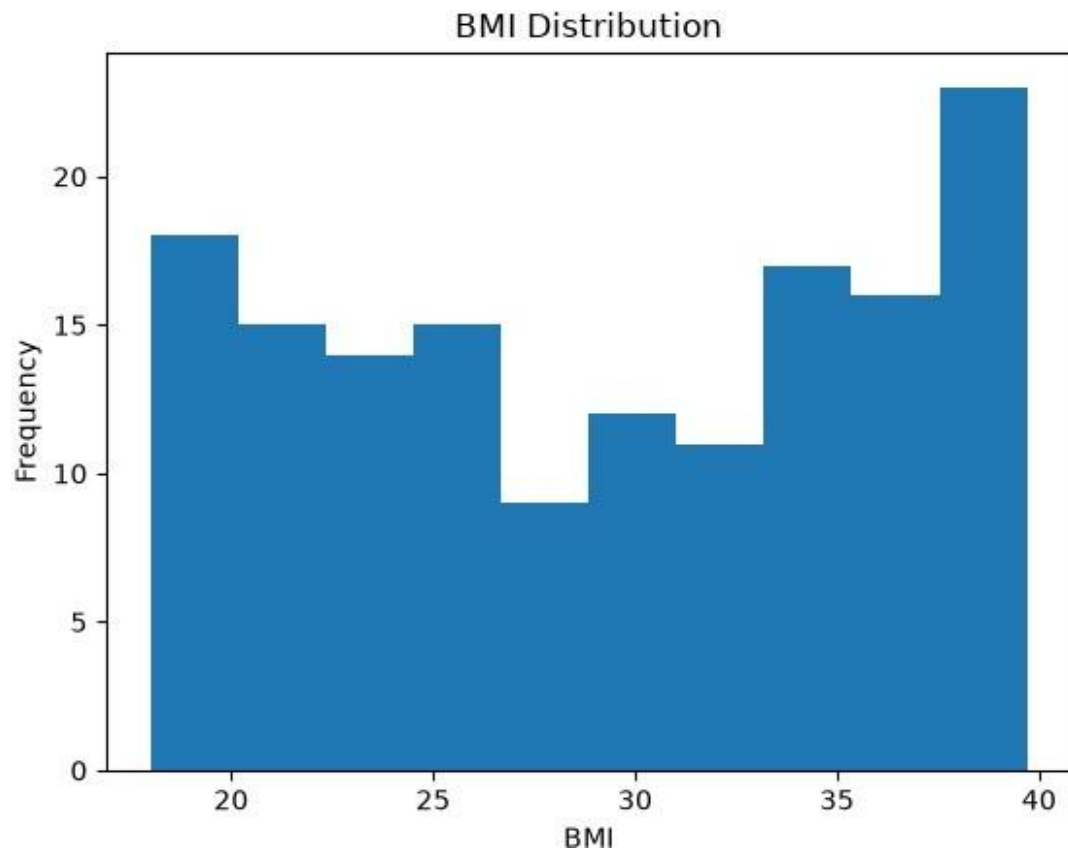
```
sns.countplot(x="Disease", data=df)
plt.title("Disease Distribution")
plt.show()
```



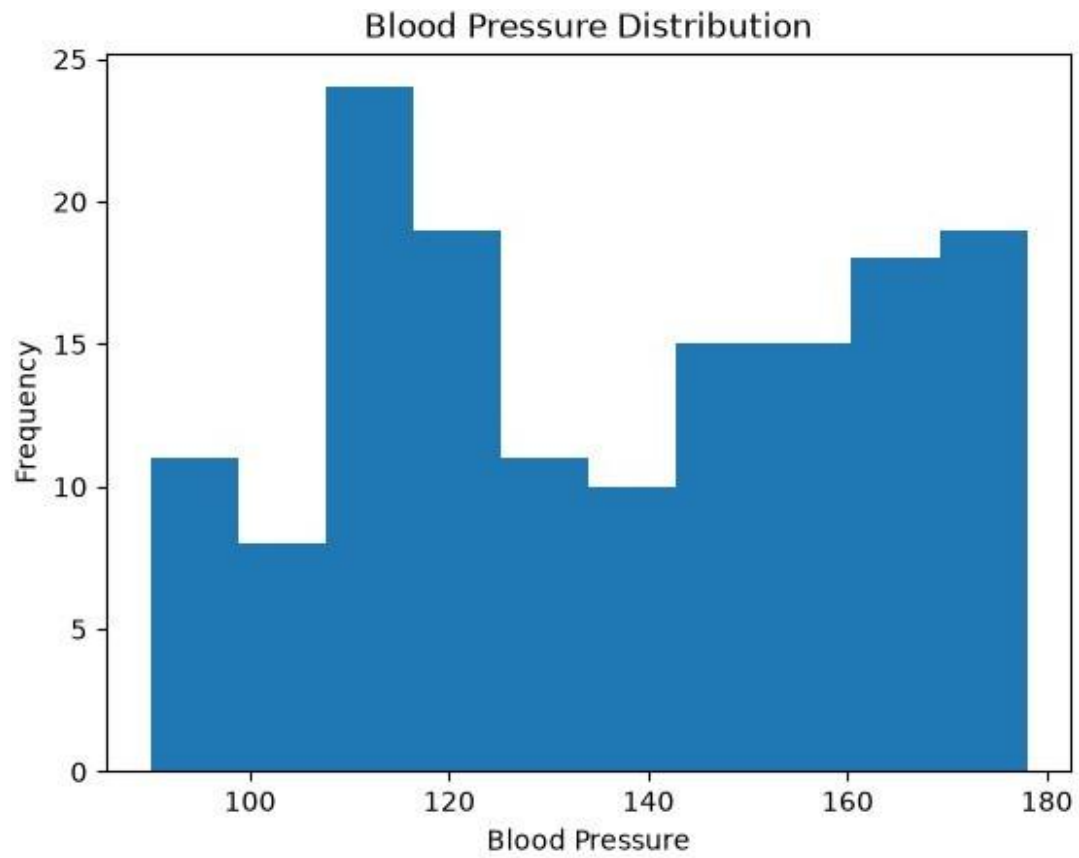
```
plt.hist(df["Age"], bins=10)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



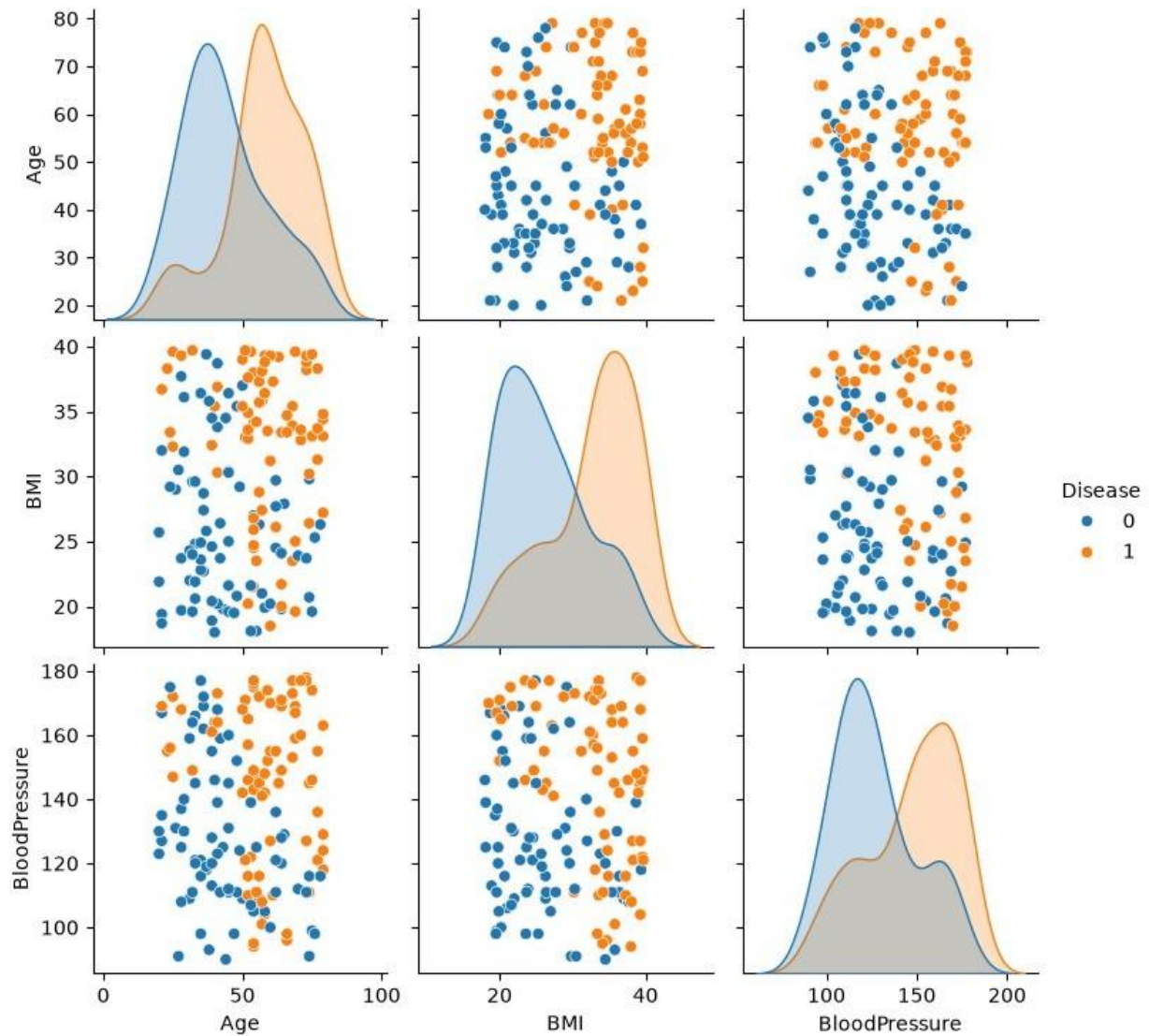
```
plt.hist(df["BMI"], bins=10)
plt.title("BMI Distribution")
plt.xlabel("BMI")
plt.ylabel("Frequency")
plt.show()
```



```
plt.hist(df["BloodPressure"], bins=10)
plt.title("Blood Pressure Distribution")
plt.xlabel("Blood Pressure")
plt.ylabel("Frequency")
plt.show()
```



```
sns.pairplot(df, hue="Disease")  
plt.show()
```



```
X = df[['Age', 'BMI', 'BloodPressure']]
```

```
y = df['Disease']
```

```
X.head()
```

	Age	BMI	BloodPressure
0	58	39.3	104
1	36	22.7	169
2	69	39.6	159
3	75	33.1	174
4	42	23.7	159

```
y.head()
```

```

0      1
1      0
2      1
3      1
4      0
Name: Disease, dtype: int64

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)

(120, 3)
(30, 3)
(120,)
(30,)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

LogisticRegression(max_iter=1000)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

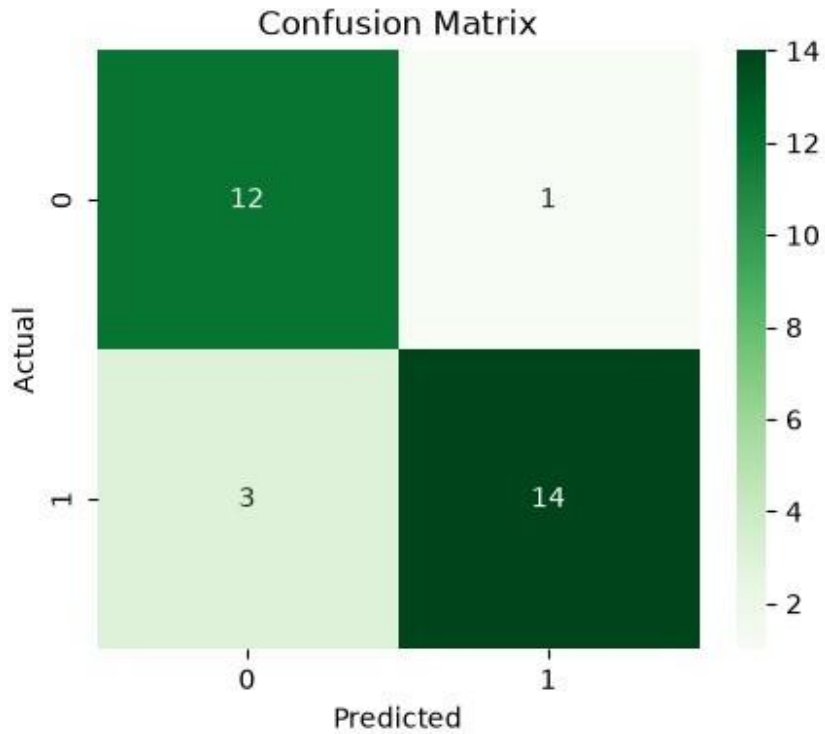
Accuracy: 0.8666666666666667

cm = confusion_matrix(y_test, y_pred)
print(cm)

[[12  1]
 [ 3 14]]

plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Greens')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

```



```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.80	0.92	0.86	13
1	0.93	0.82	0.88	17
accuracy			0.87	30
macro avg	0.87	0.87	0.87	30
weighted avg	0.88	0.87	0.87	30

```
print("Coefficients:", model.coef_)
```

```
Coefficients: [[0.22197215 0.56730698 0.13115291]]
```

```
print("Intercept:", model.intercept_)
```

```
Intercept: [-45.92127564]
```

```
patient = pd.DataFrame({
    'Age': [55],
    'BMI': [32],
    'BloodPressure': [145]
})
```

```
prediction = model.predict(patient)
```

```
print("Prediction:", prediction)
```

```
Prediction: [1]
```

```
if prediction[0] == 1:  
    print("Person is susceptible to disease.")  
else:  
    print("Person is not susceptible to disease.")
```

```
Person is susceptible to disease.
```

```
probability = model.predict_proba(patient)
```

```
print("Prediction Probability:")  
print(probability)
```

```
Prediction Probability:  
[[0.03052559 0.96947441]]
```

```
results = pd.DataFrame({  
    'Actual': y_test,  
    'Predicted': y_pred  
})
```

```
results.head()
```

	Actual	Predicted
73	0	0
18	0	0
118	0	0
78	0	0
76	0	0

```
results.to_csv("Medical_Predictions.csv", index=False)
```

```
print("Predictions saved successfully.")
```

```
Predictions saved successfully.
```